

Week 2 Executive Summary

Employee Attrition Prediction using Machine Learning

Overview

Built a machine learning system to predict employee attrition and identify key drivers of employee turnover using HR data (1,470 records, 35 features).

Key Data Insights

- Attrition rate: 16% (imbalanced dataset)
- Attrition varies across departments and job roles
- Early tenure employees show higher exit rates
- Overtime and low income strongly linked to attrition

Data Preparation

- Removed irrelevant fields (EmployeeNumber, Over18, StandardHours)
- Converted Attrition into binary (0/1)
- One-Hot Encoded categorical features
- Standardized numerical features

Model Development

- Models: Logistic Regression, Random Forest, Gradient Boosting
- Evaluation: Precision, Recall, F1-Score, ROC-AUC
- Best model selected based on overall performance (not accuracy alone)

Key Drivers of Attrition

- Overtime workload
- Monthly income
- Years at company (early-stage risk)
- Work-life balance

Business Impact

- Enables early identification of employees at risk
- Supports targeted retention strategies
- Improves engagement and workload planning
- Reduces hiring and training costs

Limitation (Important Note)

Model is based on historical data only and cannot capture personal or external factors. It should be used as a **decision-support tool**, not a replacement for HR judgment.

Conclusion

Machine learning enables HR teams to shift from reactive to proactive retention strategies by identifying at-risk employees and understanding key drivers of attrition.